

AfyaSasa — Contabo VPS Deployment Guide (Detailed)

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Hospital: Jalaram Hospital (single-tenant supervised pilot)

Stack: Docker Compose — PostgreSQL 16, Redis 7, MinIO, NestJS backend, React frontend, Nginx reverse proxy

Audience: IT administrators and DevOps engineers deploying AfyaSasa for supervised clinical use

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1. Purpose and scope

This guide walks through deploying AfyaSasa on a **Contabo VPS** (or any Ubuntu 22.04 LTS VPS) for **Jalaram Hospital** as a **supervised clinical pilot**.

What this guide covers

- Full server provisioning from empty VPS to HTTPS-accessible EMR
- Production Docker Compose configuration
- TLS termination with host Nginx and Certbot
- Database and file storage backups
- Post-deploy API verification
- Security baseline for internet-facing pilot

What this guide does not cover

- Full clinical UAT (see docs/go-live-checklist.md)
- SHA/HIE national integration (future phase)
- Multi-hospital SaaS tenancy (single-hospital Jalaram mode only)
- Load testing at scale

Important pilot principle

Do **not** enter real patient data until:

1. HTTPS is working
2. Daily PostgreSQL backups are scheduled and a restore has been tested once
3. All staff passwords have been changed from defaults
4. Clinical and IT sign-off on the go-live checklist

2. Architecture overview

AfyaSasa runs as **six Docker containers** on a single VPS:

Container	Image / build	Role	Internet exposure
nginx	nginx:1.27-alpine	Public entry — serves UI and proxies /api/ to backend	Yes — via host Nginx on 443 → Docker on 127.0.0.1:8080
frontend	Built from frontend/	React single-page application	Internal Docker network only
backend	Built from backend/	NestJS REST API + WebSocket (notifications)	Internal only

postgres	postgres:16	Clinical database (schema <code>demo</code> , tenant code <code>jalaram</code>)	Internal only — never publish 5432
redis	redis:7	Cache and job queue	Internal only — never publish 6379
minio	minio/minio	S3-compatible clinical file storage	Internal only — never publish 9000/9001

Traffic flow (production)

...

Browser (Kenya)

|

▼ HTTPS :443

Host Nginx (TLS termination, Certbot)

|

▼ HTTP 127.0.0.1:8080

Docker Nginx container

└─ / → frontend:80 (React UI)

└─ /api/ → backend:3000/api/

└─ /socket.io/ → backend:3000/socket.io/ (real-time notifications)

...

Production compose files

Always use **both** compose files together:

```
```bash
```

```
docker compose -f docker-compose.yml -f docker-compose.prod.yml up -d --build
```

```
```
```

`docker-compose.prod.yml` removes public ports from Postgres, Redis, MinIO, backend, and frontend. It binds Docker Nginx to `127.0.0.1:8080:80` so only the host reverse proxy is internet-facing.

Recommended directory layout on VPS

...

`/opt/afyasasa/` ← application root (git clone)

└─ `.env` ← production secrets (never commit)

└─ `backups/` ← PostgreSQL dumps

```

└─ docker-compose.yml
└─ docker-compose.prod.yml
└─ backend/
└─ frontend/
└─ infra/nginx.conf
└─ ops/ ← backup, test, reset scripts
...

```

3. Prerequisites checklist

Complete before ordering the VPS or starting deployment.

| # | Item | Owner | Done |
|----|--|---------------|--------------------------|
| 1 | Domain or subdomain decided (e.g. <code>emr.jalaram.co.ke</code>) | IT | ☐ |
| 2 | DNS access to create A record | IT | ☐ |
| 3 | Git repository access or release tarball | Dev | ☐ |
| 4 | Contabo account (or alternative VPS provider) | IT | ☐ |
| 5 | SSH key pair for server access | IT | ☐ |
| 6 | Strong passwords plan (Postgres, JWT, MinIO) | IT | ☐ |
| 7 | Celcom SMS keys (optional — stub mode OK for pilot) | IT | ☐ |
| 8 | Staff account list for Jalaram (<code>@jalaram.co.ke</code>) | Admin | ☐ |
| 9 | Go-live checklist printed or shared | Clinical lead | ☐ |
| 10 | Backup storage location (same VPS + offsite copy plan) | IT | ☐ |

Minimum skills required

- SSH and basic Linux administration
- Editing text files (`nano` / `vim`)
- Understanding of Docker Compose
- Basic Nginx configuration

4. Contabo VPS selection and ordering

Recommended specifications

| Resource | Minimum | Recommended (pilot) | Notes |
|----------|------------------|---------------------|---|
| RAM | 4 GB | 8 GB | Postgres + MinIO + builds need headroom |
| vCPU | 2 | 4 | Faster Docker builds and concurrent users |
| SSD | 80 GB | 160 GB | Room for DB growth, MinIO files, backups |
| OS | Ubuntu 22.04 LTS | Ubuntu 22.04 LTS | Long-term support |
| Location | EU (Germany) | Measure latency | Kenya users may prefer local VPS for production |

Contabo is suitable for **pilot and staging**. After deployment, measure page load times from Jalaram's network. If latency is unacceptable, consider a Kenya-region VPS (e.g. Truehost) using this same guide.

Contabo-specific steps

1. Log in to [Contabo Customer Control Panel](https://my.contabo.com/)
2. Order **VPS** with Ubuntu 22.04
3. Note the **public IPv4 address**
4. Set root password or upload SSH public key during setup
5. Optional: enable Contabo **external firewall** — allow only ports 22, 80, 443

After VPS is provisioned

```
""bash
ssh root@YOUR_VPS_IP
""
```

Verify connectivity and note the IP for DNS.

5. Domain name and DNS

Example domain

Use a dedicated subdomain for the EMR:

- `emr.jalaram.co.ke` (recommended)
- or `afya.jalaramhospital.com`

DNS records

| Type | Name | Value | TTL |
|------------------|---------|-------------------|----------|
| A | emr | YOUR_VPS_IP | 300-3600 |
| CNAME (optional) | www.emr | emr.jalaram.co.ke | 3600 |

Verify propagation

```
``bash
```

```
dig +short emr.jalaram.co.ke
```

Must return YOUR_VPS_IP

```
``
```

Wait 5–60 minutes after creating the record. Certbot will fail if DNS does not resolve.

6. Initial server setup

6.1 Create deploy user

SSH as root, then:

```
``bash
```

```
adduser afyasasa
```

```
usermod -aG sudo afyasasa
```

```
mkdir -p /home/afyasasa/.ssh
```

```
cp ~/.ssh/authorized_keys /home/afyasasa/.ssh/
```

```
chown -R afyasasa:afyasasa /home/afyasasa/.ssh
```

```
chmod 700 /home/afyasasa/.ssh
```

```
chmod 600 /home/afyasasa/.ssh/authorized_keys
```

```
``
```

Log out and reconnect as the deploy user:

```
``bash
```

```
ssh afyasasa@YOUR_VPS_IP
```

```
``
```

6.2 System updates and essential packages

```
``bash

sudo apt update && sudo apt upgrade -y

sudo apt install -y git curl ufw fail2ban htop unzip

...

```

6.3 Firewall (UFW)

```
``bash

sudo ufw allow OpenSSH

sudo ufw allow 80/tcp

sudo ufw allow 443/tcp

sudo ufw enable

sudo ufw status

...

```

Do not open 5432, 6379, 3000, 8080, 9000, or 9001 to the public internet.

6.4 SSH hardening (after key login confirmed)

```
``bash

sudo nano /etc/ssh/sshd_config

...

Set:

...

PasswordAuthentication no

PermitRootLogin prohibit-password

...

Then:

``bash

sudo systemctl reload sshd

...

```

6.5 Fail2ban

Installed above. Default jails protect SSH. Verify:

```
``bash

```

```
sudo systemctl status fail2ban
```

```
...
```

6.6 Set timezone (optional but recommended)

```
``bash
```

```
sudo timedatectl set-timezone Africa/Nairobi
```

```
timedatectl
```

```
...
```

7. Install Docker

```
``bash
```

```
curl -fsSL https://get.docker.com | sh
```

```
sudo usermod -aG docker $USER
```

```
newgrp docker
```

```
docker compose version
```

```
...
```

Expected output includes `Docker Compose version v2.x`.

Enable Docker on boot

```
``bash
```

```
sudo systemctl enable docker
```

```
sudo systemctl start docker
```

```
...
```

8. Deploy application code

8.1 Create application directory

```
``bash
```

```
sudo mkdir -p /opt/afyasasa
```

```
sudo chown $USER:$USER /opt/afyasasa
```

```
cd /opt/afyasasa
```

```
...
```

8.2 Clone repository

Replace with your actual repository URL:

```
``bash
git clone https://github.com/YOUR_ORG/Afya-Sasa.git .
...
```

Or upload a release tarball:

```
``bash
```

From your laptop:

```
scp afyasasa-release.tar.gz afyasasa@YOUR_VPS_IP:/opt/afyasasa/
```

On VPS:

```
cd /opt/afyasasa && tar xzf afyasasa-release.tar.gz --strip-components=1
...
```

8.3 Create backup directory

```
``bash
mkdir -p /opt/afyasasa/backups
...
```

9. Production environment configuration

9.1 Create `.env` from template

```
``bash
cd /opt/afyasasa
cp .env.example .env
nano .env
...
```

Never commit `.env` to git.

9.2 Required production values

Copy and customize the block below. Generate secrets with `openssl rand -base64 48`.

```
```env
```

## --- Application ---

---

NODE\_ENV=production

PORT=3000

FRONTEND\_ORIGIN=https://emr.jalaram.co.ke

APP\_PUBLIC\_URL=https://emr.jalaram.co.ke

## --- PostgreSQL ---

---

POSTGRES\_DB=afyasasa

POSTGRES\_USER=afyasasa

POSTGRES\_PASSWORD=<GENERATE-strong-password>

POSTGRES\_HOST=postgres

POSTGRES\_PORT=5432

DEFAULT\_TENANT\_SCHEMA=demo

DEFAULT\_TENANT\_CODE=jalaram

HOSPITAL\_NUMBER\_PREFIX=JH

TYPEORM\_MIGRATIONS\_RUN=true

## --- Redis ---

---

REDIS\_HOST=redis

REDIS\_PORT=6379

## --- JWT (generate unique values) ---

---

JWT\_ACCESS\_SECRET=<GENERATE-48-byte-secret>

JWT\_REFRESH\_SECRET=<GENERATE-different-48-byte-secret>

JWT\_ACCESS\_TTL=15m

JWT\_REFRESH\_TTL=7d

## --- MinIO / S3 (internal Docker hostnames) ---

---

MINIO\_ROOT\_USER=afyasasa

MINIO\_ROOT\_PASSWORD=<GENERATE-strong-password>

S3\_ENDPOINT=http://minio:9000

S3\_ACCESS\_KEY\_ID=afyasasa

S3\_SECRET\_ACCESS\_KEY=<same-as-MINIO-or-separate-key>

S3\_BUCKET=afyasasa-clinical-files

S3\_REGION=us-east-1

S3\_FORCE\_PATH\_STYLE=true

## --- SMS (stub until Celcom keys approved) ---

---

SMS\_PROVIDER=stub

SMS\_SENDER\_NAME=JALARAM

## --- Email (optional — leave blank until SMTP ready) ---

---

SMTP\_HOST=

SMTP\_PORT=587

SMTP\_USER=

SMTP\_PASSWORD=

SMTP\_FROM=noreply@jalaram.co.ke

## --- Frontend build args (used by Docker Compose build) ---

---

VITE\_DEFAULT\_TENANT=jalaram

VITE\_HIDE\_TENANT\_SELECTOR=true

VITE\_API\_BASE\_URL=/api/v1

...

### 9.3 Variable reference

Variable	Purpose
FRONTEND_ORIGIN	CORS and WebSocket allowed origin — must match exact browser URL including https://
APP_PUBLIC_URL	Links in emails and system messages

<code>DEFAULT_TENANT_CODE=jalaram</code>	Single-hospital mode; login hides tenant selector
<code>DEFAULT_TENANT_SCHEMA=demo</code>	Internal PostgreSQL schema name (legacy; tenant code is <code>jalar</code> am )
<code>TYPEORM_MIGRATIONS_RUN=true</code>	Auto-run DB migrations on backend startup
<code>VITE_HIDE_TENANT_SELECTOR=true</code>	Login screen does not ask for hospital code
<code>SMS_PROVIDER=stub</code>	SMS logged but not sent until Celcom keys configured

## 9.4 Generate secrets

```
```bash
```

```
openssl rand -base64 48 # run once per secret needed
```

```
```
```

Use a **different** value for each of: `POSTGRES_PASSWORD` , `JWT_ACCESS_SECRET` , `JWT_REFRESH_SECRET` , `MINIO_ROOT_PASSWORD` , `S3_SECRET_ACCESS_KEY` .

## 9.5 Pre-deploy validation (optional, on VPS or CI)

If Node.js/npm is installed on the VPS:

```
```bash
```

```
cd /opt/afyasasa
```

```
npm run preflight:prod
```

```
```
```

This checks Docker, `.env` , builds, and runs a smoke test. On a minimal VPS you may skip the npm build steps and proceed directly to Docker Compose.

# 10. Build and start the stack

From `/opt/afyasasa` :

```
```bash
```

```
docker compose -f docker-compose.yml -f docker-compose.prod.yml up -d --build
```

```
```
```

First build may take **10–20 minutes** depending on VPS CPU and network.

## 10.1 Watch startup logs

```
```bash
```

```
docker compose -f docker-compose.yml -f docker-compose.prod.yml logs -f backend
```

```
...
```

Wait for:

```
...
```

Nest application successfully started

```
...
```

Migrations run automatically when `TYPEORM_MIGRATIONS_RUN=true`.

10.2 Verify all containers

```
```bash
```

```
docker compose ps
```

```
...
```

All services should show `running` (minio-init shows `exited` with code 0 — that is normal).

## 10.3 Health check (before TLS)

Docker Nginx listens on **127.0.0.1:8080** only:

```
```bash
```

```
curl -fsS http://127.0.0.1:8080/api/v1/health
```

```
...
```

Expected:

```
```json
```

```
{"status":"ok","service":"afyasasa-backend"}
```

```
...
```

Open in browser (HTTP, temporary):

```
...
```

```
http://YOUR_VPS_IP:8080/
```

```
...
```

Port 8080 is bound to localhost only, so use SSH tunnel for remote testing before TLS:

```
```bash
```

From your laptop:

```
ssh -L 8080:127.0.0.1:8080 afyasasa@YOUR_VPS_IP
```

Then open <http://localhost:8080>

...

11. Verify migrations and tenant

11.1 Confirm Jalaram tenant exists

```
``bash
```

```
docker compose exec postgres psql -U afyasasa -d afyasasa -c \
```

```
"SELECT code, name, active FROM public.tenants WHERE code = 'jalaram';"
```

...

Expected: one row with `jalaram`, active.

11.2 Confirm admin user exists

```
``bash
```

```
docker compose exec postgres psql -U afyasasa -d afyasasa -c \
```

```
"SELECT email, active, force_password_change FROM demo.users WHERE email =  
'it@jalaram.co.ke';"
```

...

11.3 Confirm migrations table

```
``bash
```

```
docker compose exec postgres psql -U afyasasa -d afyasasa -c \
```

```
"SELECT COUNT(*) FROM public.migrations;"
```

...

Count should be greater than zero after first successful startup.

12. HTTPS with Let's Encrypt

Production must use HTTPS. Use **host Nginx** as TLS terminator; Docker Nginx stays on `127.0.0.1:8080`.

12.1 Install host Nginx and Certbot

```
``bash
```

```
sudo apt install -y nginx certbot python3-certbot-nginx
```

```
...
```

12.2 Create Nginx site configuration

```
```bash
```

```
sudo nano /etc/nginx/sites-available/afyasasa
```

```
...
```

Paste the following (replace `emr.jalaram.co.ke` with your domain):

```
```nginx
```

```
server {
```

```
listen 80;
```

```
server_name emr.jalaram.co.ke;
```

```
location / {
```

```
proxy_pass http://127.0.0.1:8080;
```

```
proxy_set_header Host $host;
```

```
proxy_set_header X-Real-IP $remote_addr;
```

```
proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
```

```
proxy_set_header X-Forwarded-Proto $scheme;
```

```
}
```

```
location /socket.io/ {
```

```
proxy_pass http://127.0.0.1:8080/socket.io/;
```

```
proxy_http_version 1.1;
```

```
proxy_set_header Upgrade $http_upgrade;
```

```
proxy_set_header Connection "upgrade";
```

```
proxy_set_header Host $host;
```

```
proxy_set_header X-Real-IP $remote_addr;
```

```
proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
```

```
proxy_set_header X-Forwarded-Proto $scheme;
```

```
proxy_read_timeout 86400;
```

```
}
```

```
}
```

```
...
```

Enable the site:

```
```bash
sudo ln -sf /etc/nginx/sites-available/afyasasa /etc/nginx/sites-enabled/
sudo rm -f /etc/nginx/sites-enabled/default
sudo nginx -t
sudo systemctl reload nginx
```
```

12.3 Obtain TLS certificate

```
```bash
sudo certbot --nginx -d emr.jalaram.co.ke
```
```

Follow prompts. Choose redirect HTTP to HTTPS when asked.

Certbot auto-renewal is installed via systemd timer:

```
```bash
sudo systemctl status certbot.timer
```
```

12.4 Update `.env` for HTTPS

Ensure these match your live URL:

```
```env
FRONTEND_ORIGIN=https://emr.jalaram.co.ke
APP_PUBLIC_URL=https://emr.jalaram.co.ke
```
```

Rebuild frontend if you changed origin after first build:

```
```bash
cd /opt/afyasasa
docker compose -f docker-compose.yml -f docker-compose.prod.yml up -d --build frontend nginx
```
```

12.5 Verify HTTPS

```
```bash
curl -fsS https://emr.jalaram.co.ke/api/v1/health
```
```

...

Open `https://emr.jalaram.co.ke` in a browser — login page should show Jalaram branding.

13. First login and account hardening

13.1 Default administrator credentials

| Field | Value |
|------------------|---|
| URL | <code>https://emr.jalaram.co.ke</code> |
| Email | <code>it@jalaram.co.ke</code> |
| Initial password | <code>ChangeMe123!</code> |
| Tenant code | Hidden — defaults to <code>jalaram</code> |

You will be prompted to change the password on first login.

13.2 Reset admin password (if login fails)

From `/opt/afyasasa` on the VPS:

```
``bash
npm run db:reset-admin
...
```

Or directly:

```
``bash
bash ops/reset-demo-admin.sh
...
```

This resets `it@jalaram.co.ke` to `ChangeMe123!` and clears lockouts.

13.3 Reset all Jalaram staff passwords

```
``bash
npm run db:reset-staff
...
```

Resets all active `@jalaram.co.ke` accounts to `ChangeMe123!` with forced change on login.

13.4 Post-login hardening tasks

| Task | Where |
|------|-------|
|------|-------|

| | |
|-------------------------------------|--|
| Change admin password | Login prompt / Account Center |
| Verify staff accounts | Admin → Users |
| Disable unused demo accounts | Admin → Users |
| Confirm hospital name/logo on login | Login screen (from database) |
| Enable Pharmacy module if needed | Hospital Control Center → Modules |
| Do not expose Swagger publicly | Block <code>/docs</code> at host Nginx or IP allowlist |

13.5 Restrict Swagger in production (recommended)

Add to host Nginx HTTPS server block:

```
```nginx
```

```
location /docs {
```

```
deny all;
```

```
return 404;
```

```
}
```

```
```
```

Reload Nginx after change.

14. Post-deploy testing

14.1 Smoke test (from VPS)

```
```bash
```

```
cd /opt/afyasasa
```

```
BACKEND_URL=http://127.0.0.1:8080/api/v1 \
```

```
FRONTEND_URL=http://127.0.0.1:8080 \
```

```
TENANT=jalaram \
```

```
bash ops/smoke-test.sh
```

```
```
```

Note: smoke test also probes Swagger on port 3000 internally — in production the important checks are health and login via port 8080.

14.2 Full workflow test suite

Run all API acceptance tests:

```
```bash
cd /opt/afyasasa
API=http://127.0.0.1:8080/api/v1 TENANT=jalaram npm run test:workflows
```
```

This runs:

| Script | Workflow |
|--------------------------------|---------------------------------|
| ops/smoke-test.sh | Health + login |
| ops/opd-workflow-test.sh | OPD check-in → triage → consult |
| ops/ipd-workflow-test.sh | Admission → discharge |
| ops/lab-workflow-test.sh | Lab order → result → review |
| ops/radiology-workflow-test.sh | Imaging order → report |
| ops/inventory-workflow-test.sh | Stock, requisition, transfer |
| ops/pharmacy-workflow-test.sh | Prescribe → dispense |

All scripts must pass before inviting clinical staff.

14.3 External HTTPS test (from laptop)

```
```bash
curl -fsS https://emr.jalaram.co.ke/api/v1/health
```
```

Manual UI walkthrough: see `docs/core-workflow-test-plan.md` .

15. Backups and disaster recovery

15.1 PostgreSQL backup (daily)

Use the project script:

```
```bash
cd /opt/afyasasa
npm run db:backup
```

**Writes to backups/afyasasa-YYYYMMDD-HHMMSS.sql**

...

Or with gzip for cron:

```
```bash
#!/usr/bin/env bash

set -euo pipefail

cd /opt/afyasasa

STAMP=$(date -u +%Y%m%dT%H%M%SZ)

docker compose exec -T postgres pg_dump \
-U afyasasa afyasasa --clean --if-exists \
| gzip > "backups/afyasasa-${STAMP}.sql.gz"

find backups -name '*.sql.gz' -mtime +14 -delete

echo "Backup complete: backups/afyasasa-${STAMP}.sql.gz"
```
```

...

Save as `/opt/afyasasa/ops/cron-backup.sh` , `chmod +x`.

## 15.2 Schedule daily backup

```
```bash
crontab -e
...

Add:
...

0 2 * /opt/afyasasa/ops/cron-backup.sh >> /opt/afyasasa/backups/backup.log 2>&1
...
```
```

## 15.3 Restore test (mandatory once before live data)

```
```bash
cd /opt/afyasasa

gunzip -c backups/afyasasa-YYYYMMDDTHHMMSSZ.sql.gz | \
docker compose exec -T postgres psql -U afyasasa -d afyasasa
...
```
```

Or for uncompressed dump:

```
```bash
```

```
npm run db:restore -- backups/afyasasa-YYYYMMDD-HHMMSS.sql
```

```
...
```

Verify login and a sample patient record after restore.

15.4 MinIO / clinical files backup

MinIO data lives in Docker volume `afyasasa_minio-data`.

List volumes:

```
```bash
```

```
docker volume ls | grep minio
```

```
...
```

Backup options:

1. **Volume archive:** `docker run --rm -v afyasasa_minio-data:/data -v /opt/afyasasa/backups:/backup alpine tar czf /backup/minio-$(date +%Y%m%d).tar.gz -C /data .`
2. **MinIO client mirror** to external S3 or another server (install `mc` on host)

Schedule weekly MinIO backups at minimum.

## 15.5 Offsite copies

Copy `backups/*.sql.gz` and MinIO archives to:

- Separate cloud storage (Google Drive, S3, Backblaze)
- Second server
- Encrypted external drive (weekly physical rotation)

**A backup that exists only on the same VPS as production is not disaster recovery.**

# 16. Updates, redeploy, and rollback

## 16.1 Standard update procedure

```
```bash
```

```
cd /opt/afyasasa
```

```
git pull
```

```
docker compose -f docker-compose.yml -f docker-compose.prod.yml up -d --build
```

```
docker compose logs -f backend --tail 80
```

```
...
```

Migrations apply automatically on backend restart.

16.2 Post-update verification

```
```bash
API=http://127.0.0.1:8080/api/v1 TENANT=jalaram npm run test:workflows
```
```

16.3 Rollback if update fails

- 1. Stop stack: `docker compose -f docker-compose.yml -f docker-compose.prod.yml down`
- 2. Checkout previous git tag/commit: `git checkout <previous-tag>`
- 3. Restore database from last good backup if migrations broke schema
- 4. Rebuild and start: `docker compose -f docker-compose.yml -f docker-compose.prod.yml up -d --build`

Always take a backup **before** major updates.

17. Monitoring and maintenance

17.1 Daily checks

| Check | Command | | |
|--------------------|---|------------------------------|------------------------|
| Containers running | <code>docker compose ps</code> | | |
| API health | <code>curl -fsS https://emr.jalaram.co.ke/api/v1/health</code> | | |
| Disk space | <code>df -h</code> | | |
| Memory | <code>free -h</code> | | |
| Recent errors | <code>`docker compose logs backend --since 24h 2>&1 \</code> | <code>grep -i error \</code> | <code>tail -20`</code> |
| Backup file exists | <code>ls -lh /opt/afyasasa/backups/</code> | | |

17.2 Docker log rotation

Create `/etc/docker/daemon.json` :

```
```json
{
 "log-driver": "json-file",
 "log-opts": {
 "max-size": "50m",
```

```
"max-file": "3"
}
}
...
```

Then: `sudo systemctl restart docker` (brief downtime — schedule maintenance window).

### 17.3 Auto-start on reboot

Docker service is enabled via systemd. Containers use `restart: unless-stopped` if added to compose; verify after reboot:

```
``bash
sudo reboot
```

## After reconnect:

---

```
docker compose ps
curl -fsS http://127.0.0.1:8080/api/v1/health
...
```

## 18. Celcom SMS integration

When Celcom Africa approves your API keys:

```
``env
SMS_PROVIDER=celcom_africa
CELCOM_API_KEY=<from Celcom>
CELCOM_PARTNER_ID=<from Celcom>
CELCOM_SHORTCODE=JALARAMHNKR
...
```

Restart backend:

```
``bash
docker compose -f docker-compose.yml -f docker-compose.prod.yml up -d backend
...
```

Test a single SMS from Admin → Notifications before enabling appointment reminders or bulk sends.

---

## 19. Security hardening

### Network

- Only ports **22, 80, 443** open on UFW and Contabo firewall
- Postgres, Redis, MinIO, backend **not** published to `0.0.0.0`
- MinIO console (9001) **not** exposed in production compose

### Secrets

- Rotate all defaults from `.env.example` before go-live
- Store `.env` with permissions `chmod 600 .env`
- Never commit `.env` or paste secrets in chat/email

### Application

- RBAC enforced on backend — frontend nav hiding is not security
- Force password change for all staff after reset
- Pilot under **supervised** clinical use until UAT sign-off

### SSH

- Key-based auth only (disable password auth)
- Consider changing SSH port (optional)
- fail2ban active

### TLS

- Certbot auto-renewal enabled
- HSTS can be added after stable HTTPS (optional advanced step)

## 20. Troubleshooting reference

Problem	Likely cause	Fix
502 Bad Gateway	Backend not running	<code>docker compose ps ; docker compose logs backend</code>
502 after TLS	Host Nginx wrong upstream	Confirm <code>proxy_pass http://127.0.0.1:8080</code>
Migrations fail	Wrong Postgres password	Match <code>.env</code> to volume; or recreate volume (data loss)
Login fails	Wrong password / lockout	<code>npm run db:reset-admin</code>

Login fails	Wrong tenant	Ensure <code>VITE_DEFAULT_TENANT=jalaram</code> and rebuild frontend
Blank white page	Frontend build error	<code>docker compose ... up -d --build frontend</code>
CORS errors	Origin mismatch	<code>FRONTEND_ORIGIN</code> must exactly match browser URL
WebSocket / notifications dead	Missing upgrade headers	Add <code>/socket.io/</code> block to host Nginx (Section 12.2)
Certbot fails	DNS not propagated	<code>dig +short your.domain ; wait and retry</code>
Slow from Kenya	EU VPS latency	Measure; consider Kenya-region VPS
<code>npm run deploy:prod</code> issues	Wrong compose file	Use both files: <code>-f docker-compose.yml -f docker-compose.prod.yml</code>
Out of disk	Logs or backups	<code>df -h ; prune: docker system prune -a</code> (careful)

Useful diagnostic commands

```bash

Container logs

`docker compose logs backend --tail 100`

`docker compose logs nginx --tail 50`

Enter Postgres

`docker compose exec postgres psql -U afyasasa -d afyasasa`

Test login via API

```
curl -sS -X POST http://127.0.0.1:8080/api/v1/auth/login \
-H "Content-Type: application/json" \
-H "X-Tenant: jalaram" \
-d '{"email":"it@jalaram.co.ke","password":"ChangeMe123!","device":"curl"}'
...

```

21. Post-deploy sign-off checklist

| # | Item | Pass |
|---|------|------|
|---|------|------|

| | | |
|----|---|--------------------------|
| 1 | HTTPS works in browser | ☐ |
| 2 | Login page shows Jalaram hospital name | ☐ |
| 3 | Admin login + password change works | ☐ |
| 4 | <code>npm run test:workflows</code> all green | ☐ |
| 5 | Daily Postgres backup scheduled | ☐ |
| 6 | Restore test completed once | ☐ |
| 7 | MinIO backup plan documented | ☐ |
| 8 | Staff passwords changed from default | ☐ |
| 9 | Swagger not publicly accessible | ☐ |
| 10 | Clinical UAT started (<code>docs/go-live-checklist.md</code>) | ☐ |

22. Quick command reference

```
```bash
```

### Start (production)

```
cd /opt/afyasasa
```

```
docker compose -f docker-compose.yml -f docker-compose.prod.yml up -d --build
```

### Stop

```
docker compose -f docker-compose.yml -f docker-compose.prod.yml down
```

### Logs

```
docker compose logs -f backend
```

### Health (local)

```
curl -fsS http://127.0.0.1:8080/api/v1/health
```

### Health (public)

```
curl -fsS https://emr.jalaram.co.ke/api/v1/health
```

# Smoke test

BACKEND\_URL=http://127.0.0.1:8080/api/v1 FRONTEND\_URL=http://127.0.0.1:8080  
TENANT=jalaram bash ops/smoke-test.sh

# Full workflow tests

API=http://127.0.0.1:8080/api/v1 TENANT=jalaram npm run test:workflows

# Backup

npm run db:backup

# Reset admin

npm run db:reset-admin

...

**Pilot URL:** https://emr.jalaram.co.ke (replace with your domain)

**Admin:** it@jalaram.co.ke / ChangeMe123! (change immediately)

**Tenant:** jalaram (hidden on login)

## 23. Related documents

Document	Purpose
docs/go-live-checklist.md	Full clinical and IT sign-off
docs/core-workflow-test-plan.md	Manual UAT scenarios by role
docs/core-workflow-gap-matrix.md	Feature completeness matrix
docs/go-live-readiness-directive.md	Master readiness plan
docs/local-testing-guide.md	Local development testing
docs/inventory-architecture.md	Pharmacy and store design
docs/contabo-deployment-guide.md	Short deployment summary

End of document — AfyaSasa Contabo Deployment Guide v1.1

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